

Day 16 Hybrid Notes

Connecting center, radius, distance, equations, and graphs | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets
I can identify center, radius, diameter, chord, tangent, and secant.
I can read and write a circle equation in center-radius form.
I can graph a circle using its center, radius, and four anchor points.

Part 1: Cloze Notes - Essential Ideas
1. A circle is the set of points the same distance from a _____.
2. A radius connects the center to a point on the _____.
3. A diameter is twice the _____.
4. A tangent touches a circle at exactly one _____.
5. A secant intersects a circle at _____.
6. Standard form is $(x-h)^2 + (y-k)^2 =$ _____.
7. The signs inside the parentheses are opposite the center _____.

Part 2: Outline Notes - Big Picture A. Circle equations come from the distance _____. B. The center controls horizontal and vertical _____. C. The right side of the equation gives radius _____. D. Graphing begins with the center and four anchor _____. _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ _____
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Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 16 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. State the center and radius of $(x-3)^2+(y+2)^2=25$.

Setup/work: _____

Answer: _____

2. Write the equation of a circle with center $(-4,1)$ and radius 6.

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student says the radius of $(x-2)^2+(y+1)^2=49$ is 49.

What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A circular sensor is centered at $(2,-1)$ and reaches the point $(8,7)$. Find its radius and equation.

Setup: _____

Answer: _____

Sentence answer: _____